



EXTENSION RESEARCH CATEGORY

PAPER TITLE	TREAT-U : Coconut Livelihood Program
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PROJECT LEADERS	Engr. Catherine B. Arroyo Engr. Kenny John P. Ballera Engr. Sandra S. Balsote Engr. Abelardo Y. Bermejo Engr. Alex D. Buenafe Engr. Bonifacio A. Hontiveros Engr. Francis Belle D. Lampong Mr. Jonie B. Lungsod
TRAINING PROGRAM	Technology from Research for Extension, Adoption, Transfer and Utilization (TREAT-U) Program; CAPSU-DAR Coconut Livelihood Program
CAMPUS	Pontevedra
CAMPUS EXTENSION PROGRAM	Kabuy-anan, Ika-ayong Lawas, Panguma, Ihibalo kag Teknolohiya (KALAPIT)
BSABE BANNER PROGRAM	Alamin at Linangin ang Ating Yaman para sa Kaunlaran (ALAY–Kaunlaran)
BENEFICIARIES	Agrarian Reform Cooperative Coconut Farmers
TIME FRAME/DURATION	3 years

RATIONALE

The "coconut tree," scientifically known as *Cocos nucifera* (L.), is the most naturally widespread fruit plant on Earth (National Library of Medicine, 2015). The Philippines, after Indonesia, is the world's second largest producer of coconut products. The Philippines' Davao Region is the leading coconut producer, accounting for 14.4% of the country's total coconut production in 2018. As a result, the coconut industry is one of the most important sources of employment in the Philippines (Philippine Coconut Industry, 2022).

With so many uses for every part of the coconut, there's no wonder why the coconut palm tree is known as "the tree of life". It provides basic necessities for humans. The 10 most common uses of coconut are (1) The Flesh: Food, Milk, and Flour; (2) The Water: A Healthy, Refreshing Drink; (3) The Oil: For Cooking, Skin, and Hair; (4) The Shells: To Steam Food and Craft With; (5) The Husks: A Natural Scrubber and Craft Material; (6) The Husks: Ropes; (7) Tree Leaves: Thatching; (8) The Sticks: Brooms; (9) The Wood: Fires in Traditional Kitchens; and (10) The Flowers: Medicine (McCormick, 2017).

Acknowledging the importance of coconut to people, the Department of Agricultural and Biosystems Engineering (ABE) at Capiz State University (CAPSU), Pontevedra Campus has developed a number of mature technologies as products of research. These research products were also based from the needs of the community. A number of these have pending patent/utility model applications. Some of these have gained popularity over exhibits and to government officials like the Department of Agriculture and Philippine Coconut Authority. A lot of people and organizations have expressed interest to own the department's machines and have been inquiring how to acquire them.

The commercialization of technologies for coconut is part of the department's extension program, **TREAT-U**, which is short for **Technology** from **Research** for **Extension, Adoption, Transfer and Utilization**. It sounds like "I'll treat you" or the Tagalog translation "i-treat kita". To treat is "to give something for pleasure or delight". The letter **U** which sounds like the pronoun "you" refers to the various clienteles of the university. This is the essence of the program which is the giving or distribution of machines for the clienteles' adoption or utilization. Moreover, one of CAPSU RDE's objectives is to package and disseminate generated/verified R&D information for possible adoption, utilization and commercialization. This means that good products of research be extended to the community to fulfill the mandate and commitment of the University to improve the quality of life in the service area through meaningful RDE activities (CapSU Code).

CAPSU's idea of research-extension collaboration is cascaded down to CapSU-Pontevedra's Agricultural and Biosystems Engineering (ABE) Department with its Alamin at Linangin Ating Yaman para sa Kaunlaran (ALAY-Kaunlaran) as extension banner program. It is also a means of fulfilling the department's second objective which states "to deliver research-based and mature technologies to the identified community for increased farm productivity and profitability". After utilizing the machines, improvement of the clienteles' economic or work conditions is attained.

In the last quarter of 2022, the Department of Agrarian Reform (DAR) – Capiz expressed its willingness to own the BSABE Coconut Dehusker. The commercialization of it was finalized in the early part of 2023 where a unit was delivered to Lonoy Agrarian Reform Cooperative (LARC), the DAR beneficiary in Sapián, Capiz. ALAY – Kaunlaran, the BSABE extension banner program, does not limit its engagement with DAR to coconut dehusking only. It also considers the need to deliver to the beneficiaries other coconut-related technologies, provide technical assistance and livelihood opportunities to them. Hence, this CAPSU-DAR Capiz Coconut Livelihood Program arose.

OBJECTIVES

The objectives of the extension research are to:

1. Enumerate the extension activities in line with the Coconut Livelihood Program;
2. Draw feedbacks from the DAR beneficiaries on the conducted activities; and
3. Find out the training needs of the DAR beneficiaries.

PROGRAM DESCRIPTION

The **CAPSU-DAR Coconut Livelihood Program**, as shown in Figure 3, is in line with the department's banner program called ALAY-KAUNLARAN (Figure 1) and the **TREAT-U** Program (Figure 2). **ALAY-Kaunlaran** includes six (6) major programs as shown in the model. Each program has specific projects below it. These are Extension Programs No. 1 (Training and Skills

Development Support), No. 2 (Livelihood Training), No. 4 (Expert Assistance Service and Consultancy) and No. 5 (Technology Demo and Income-Generating Projects).



Figure 1. The six (6) major components of ALAY-Kaunlaran.

The **TREAT-U** Program launched in 2020 has two phases, namely:

- (1) Pre-Adoption Phase
 - (a) Technology Demonstration
 - (b) MOA Signing
 - (c) Fabrication
 - (d) Pre-Training/ Training
- (2) Adoption phase
 - (a) Turning Over
 - (b) Training and Utilization
 - (c) Supervision/Monitoring
 - (d) Impact Assessment

The concept flow of the TREAT-U Program starts with the **pre-adoption phase**. It begins with a demonstration of the prototype technology. After this activity, the participants/parties may be interested or not interested. For those interested of the technology, they will write a letter of intent addressed to the RDE of CapSU-Pontevedra through the ABE Department. Once approved, there will be a MOA signing between the interest Party and CapSU. Then fabrication of the technology starts. Since the ABE department does not have extra units aside from the prototype, the fabrication may be CapSU, LGU or NGO sponsored. Before the technology adoption, there will first be a pre-training/training on the use of the technology which includes safety, maintenance and skills development.

The second part of the **TREAT-U** Program is the **adoption phase**. It begins with the technology turning over of technology (technology transfer) to the interested party for utilization. Before using the technology, a massive training/retraining on its operation and maintenance will be conducted. While using the adopted technology, ALAY-Kaunlaran will supervise and give technical assistance such as on repair and maintenance, economic or cost and return analysis, etc. After enough utilization of the technology, ALAY-Kaunlaran will conduct impact assessment as shown in Figure 2.

The TREAT- U Program's conceptual framework will serve as a guide of CAPSU-DAR Capiz Coconut Livelihood Program whenever the activity involves a technology. The Coconut Livelihood Program's framework (Figure 3) shows the different activities to be undertaken by both parties.

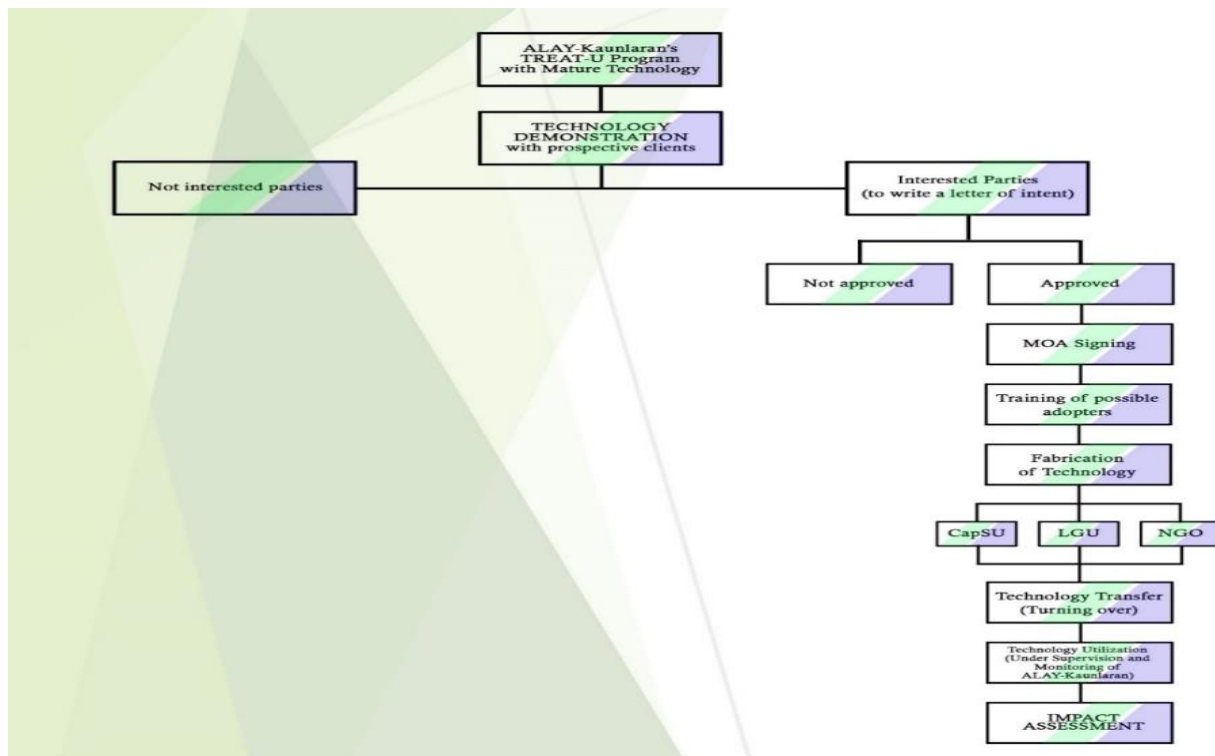


Figure 2. The concept flow of the **TREAT-U** Program



Figure 3. The framework of the CAPSU-DAR Coconut Livelihood Program.

METHODOLOGY

The conduct of the training/s will be based on the agreed CAPSU-DAR Coconut Livelihood Program, TNA or letter request. The conceptual framework of the TREAT-U Program will also be followed. It has two phases: the pre-adoption phase and the adoption phase. The pre-adoption phase consists of technology demonstration, MOA signing, fabrication and pre-training. The adoption phase starts from turning over of the equipment, technology utilization/retraining and impact assessment.

Technology Demonstration

After learning their interests on the BSABE machines for coconut, the DAR project in-charge or leaders were set for technology demonstration. Their feedbacks were noted.

When they expressed interest to own the technology, they were advised to write a letter of intent to CapSU-Pontevedra RDE through the ABE Department. A meeting was set in the presence of the RDE Central Office authorities. A second meeting was set with the fabricator.

Memorandum of Agreement (MOA) Signing

The MOA was drafted by ALAY-Kaunlaran in consultation with CapSU-RDE office. The draft copy of the MOA was provided to the interested party, RDE and higher authorities for correction or improvement. A final copy of the MOA was generated for signing of both parties. The signed MOA was notarized and distributed to the concerned party and offices.

Fabrication and Turning Over of the Technology

After everything was finalized, fabrication of technology followed. This was done by a private fabricator at Jamindan, Capiz. The monetary transaction was between DAR and the fabricator. CAPSU's participation was to see to it that the machine is fabrication is well-done in terms of materials, design and parts. After a series of pre-trials, the machine was turned over to LARC beneficiaries in the later part of January 2023.

Conduct of Training

To ensure the correct operation and maintenance of the coconut dehusker, a training was done. The faculty members of the ABE Department were the facilitators of the training/s. They were supported by the staff of the department and the RDE Office. The trainings were conducted at the DAR beneficiaries' area.

Each training had a maximum of 35 participants. The training was divided into lectures and hands-on. More time was allocated for the desired competencies and skills. The trainees were subjected to testing and evaluation on all aspects of the training. Certificate of training was given to those who successfully pass the training. The trainees were screened by the officers of Lonoy Agrarian Reform Cooperative (LARC).

Budget allocation was sought from the CapSU Administration in accordance with accounting rules and procedures. The activities were properly documented and reported.

The tools, equipment and materials were provided by the Capiz State University. The facilities of LARC used in the training program were the following:

1. Training hall with audio-visual facilities
2. Testing materials and cooking utensils

RESULTS AND INTERVENTIONS

Presented in this paper are (a) the activities done related to the CAPSU-DAR Coconut Livelihood Program in 2022-2023, (b) evaluation results and the impact of the coconut-related trainings done and (c) needs assessment of the DAR beneficiaries.

Activities Done For CAPSU-DAR Coconut Livelihood Program

Table 1 shows the activities done by CAPSU and DAR relative to the Coconut Livelihood Program. These activities conformed with the conceptual framework of TREAT-U. The CAPSU-DAR engagement started in the later part of 2022 when DAR representatives visited CAPSU-Pontevedra looking for a coconut dehusker for their beneficiaries (Picture 1). Several meetings were done by BSABE-RDE before the commercialization of the machine was done (Picture 2). During one of the BSABE-RDE-DAR meetings (Picture 3), it was agreed that the fabrication be done by a private individual at Jamindan, Capiz. The target date was last week of December 2022.

The fabricator in Jamindan, Capiz was visited by BSABE Faculty several times to ensure that the machine will be functional and meets the design criteria (Picture 4). There were pre-trials done in the presence of the beneficiaries (Pictures 5-6). The machine was finally done and turned over to Lonoy Agrarian Reform Cooperative (LARC) who are the recipients of the coconut technology. The turning over was witnessed by Dr. Honey Lee E. Casa, CAPSU-Pontevedra Campus Administrator together with Dr. Abelardo Bermejo (Extension Chair), Engr. Francis Belle Lampong (BSABE Program Coordinator), Engr. Marty Delfin (researcher and BSABE Extension Coordinator, two DAR representatives and the fabricator (Picture 7).

To make the CAPSU-DAR engagement wider, a preliminary talk between the campus Extension Department and DAR management happened in the early part of February. Both parties agreed not just to limit the engagement to coconut dehusking but to other activities related to coconut. A Coconut Livelihood Program was drafted and finalized.

Picture 8 shows the formal commencement of the CAPSU-DAR Coconut Livelihood Program on March 29, 2023 with a MOA signing attended by CAPSU and DAR representatives at the headquarters of LARC, Lonoy, Sapián, Capiz. CAPSU was represented by Dr. Abelardo Y. Bermejo (Campus Extension Chair), Engr. Francis Belle Lampong (BSABE Program Coordinator) while DAR was represented by Ms. Obierna (Project Officer) and Mr. Arostique (head of office). Also present were the Extension staff (Mr. Jiro Bangud and Mr. Loyd Altoveros), BSABE Faculty and staff (Engrs. Arroyo, Ballera, Balsote, Delfin, Hontiveros and Mr. Burbano), and LARC officers and members.

A Training on the Operation and Maintenance of the BSABE Coconut Dehusker was done on March 29, 2023 with 35 participants who are the members of LARC. The participants were taught by Engr. Ballera on Health and Safety, by Engr. Delfin on the operation of the machine and by Engr. Hontiveros on the maintenance of the machine.

To make use of the dehusked coconut, a follow-up training on the operation and maintenance of the BSABE Coconut Grater was conducted on the 30th of March, 2023. It was also attended by 35 LARC members. Engrs. Arroyo, Balsote and Lampong were the resource persons for this training.

A training on coconut utilization was done on March 31, 2023 to more or less 35 LARC members. This made use of the grated coconut for kakanin making. BSAH Faculty, Mrs. Leah Yakit and Mr. Rommel Abaldonado with their students were tapped to collaborate with BSABE. As requested by the members, they wanted to learn how to make “panucha” in banana leaves, a by-product made of grated coconut and sugar. They were also taught how to make biko with latik and kalamayhati.

Table 1. Activities related to CAPSU-DAR Coconut Livelihood Program.

TREAT-U COCONUT LIVELIHOOD PROGRAM ACTIVITY	DATE CONDUCTED	NO. OF PARTICIPANTS	NO. OF HOURS	BUDGET ALLOCATION
1) Visit of DAR to CAPSU-BSABE and technology demonstration	Nov 2022	2 DAR 5 CAPSU	2	
2) Meeting of DAR and CAPSU RDE	Nov 2022	2 DAR 10 CAPSU	2	
3) Meeting of DAR,CAPSU RDE and Fabricator	Nov 2022	2 DAR 1 Fabricator 10 CAPSU	4	
4) Visit of CAPSU to Fabricator	Dec 2022	1 Fabricator 5 CAPSU	2	
5) 1 st Pre-trial of Dehusker	Jan 2023	2 DAR 5 CAPSU 1 Fabricator	3	
6) 2 nd Pre-trial	Jan 2023	5 LARC 2 DAR 5 CAPSU	3	
7) Turning Over of Dehusker	January 31, 2023	10 LARC 2 DAR 8 CAPSU	4	
8) MOA Signing	March 29, 2023	35 LARC 10 CAPSU 3 DAR	1	
9) Training on the Operation and Maintenance of BSABE Coconut Dehusker	March 29, 2023	35 LARC 10 CAPSU	8	Php 10,272.00
10) Training on the Operation and Maintenance of BSABE Coconut Grater	March 30, 2023	35 LARC 8 CAPSU	8	Php 10,120.00
11) Training on Coconut Utilization	March 31, 2023	35 LARC 8 CAPSU	8	Php 9,210.00



Figure 4. Pictures of different activities related to the CAPSU-DAR Coconut Livelihood Program.

Training Assessment and Impact

After each training related to the CAPSU-DAR Coconut Livelihood Program on March 29-31, 2023, the participants were requested to indicate their evaluation. Surprisingly, all the three trainings got an average rating of 5.00. The participants had perfect scores on all the items in the evaluation sheet (Table 2). The evaluation items were written in the vernacular and centered around the following criteria: (a) topic – *topiko*, (b) speakers/trainers – *manunudlo o manughanas*, (c) facilitators – *manugpatigayon*, (d) food and accommodation – *pagkaon kag akomodasyon*, and (e) time management – *pag-obserbar sang tama nga oras*.

As a result of the above-mentioned trainings, LARC was able to make coconut products of their own and was able to display these products during the CAPIZTAHAN celebration in April 2023. They were able to sell and earn profit from their coconut products (Figure 5).

Table 2. Evaluation results of the three trainings conducted on March 29-31, 2023.

CRITERIA	MEAN		
	Coconut Dehusker	Coconut Grater	Coconut Utilization
TOPIKO			
1. Nagakaigo gid ang topiko sa akon kinahanglan sa pangabuhì kag trabaho.	5	5	5
2. Ang pagpasunod sang topiko maayo kag may yara nga sistema	5	5	5
3. Ang presentasyon o demo nga ginpakita nagasunod sa tinutuyo sang training	5	5	5
4. Nagahatag sang oportunidad nga makapamangkot para mas maintindihan	5	5	5
ANG MANUGTUDLO/MANUGHANAS			
1. Gapakita sang iya nahibal-an sa topiko.	5	5	5
2. Sigurado siya nga matuod kag mapatihan ang iya ginasaysay nga impormasyon.	5	5	5
3. Matawhay ang pagpasunod sang iya nga topiko agud madali maintindihan.	5	5	5
4. Maayo siya magsaysay kag wala gautay-utay ang iya ginahambal.	5	5	5
5. Makašabat siya sang pamangkot kag mapaintindi ang iya sabat sa tagapalamati.	5	5	5
6. Nagahatag sang inspirasyon sa tagapalamati agud nga mangin interesado ang topiko	5	5	5
MANUGPATIGAYON			
1. Nagapakita sang ila propesjonalismo sa ila pagbulig sa paghanas o seminar	5	5	5
2. Matinahuron, pasensyoso kag mabinuligon nga nagabuylog sa paghanas.	5	5	5
3. Nagahatag sang matawhay kag maayo nga lugar para sa pagtu-on sang bag o nga impormasyon.	5	5	5
4. Ginamanehar ang bilog nga grupo agud nga magtawhay ang dalagan sang training.	5	5	5

Table 2 continued			
PAGKAON KAG AKOMODASYON			
1. Maayo ang kalidad kag masustansya ang ila ginahatag nga pagkaon.	5	5	5
2. Ang pag-amuma nagahatag sa mga nagabuylog sang training sang katawhay	5	5	5
3. Nagakaigo kag matawhay ang lugar nga ginhiwatan sang training. (Wala sang butang nga makadisturbo sa atensyon sang mga partisipante)	5	5	5
PAG OBSERBAR SANG TAMA NGA ORAS			
1. Gilayon nga ginaaksyonan ang ginahinyo sa paghanas.	5	5	5
2. Nagakaigo gid ang akun kinahanglanon sa paghanas.	5	5	5
MEAN	5.00	5.00	5.00

The LARC trainees applied in real life the knowledge and skills they gained from the trainings conducted by CAPSU. During the April 2023 CAPIZTAHAN, they made and included in their booth the selling *kakanin* (biko and panucha). These are the results of the series of trainings they attended on March 29-31, 2023 – coconut dehusker, coconut grater and coconut utilization/product development.



Figure 5. Picture of LARC booth during the CAPIZTAHAN celebration in April 2023 where Coconut Livelihood Program products like *panucha* were displayed. In the photo is Mr. Rommel Abaldonado, one of the trainers from CAPSU-Pontevedra.

Training Needs Assessment of DAR Beneficiaries

The DAR beneficiaries at Lonoy, Sapián, Capiz were made to answer the Training Needs Assessment Form of the RDE Office. The percentages of those who need a particular training are found in Tables 3a to 3c. In Table 3a which pertains to livelihood, more than one-half (50%) of the respondents wanted a training on coco sugar production, cooking oil making, virgin coconut making, handicraft/souvenir items making, vegetable production, kakanin making and vinegar making, chicken production, swine production, organic farming, baking, cooking and cookery, door mar/rag making, food processing and pasta noodles making. It was noted that among these, three are related to the Coconut Livelihood Program – coconut sugar, virgin coconut and cooking oil making.

Among the trainings under the category Education and Humanities (Table 3b), majority of them indicated no interest to almost all of the items except for the training on customer service. In Table 3c where the items were categorized as Community Needs and Technology, the greatest number of respondents indicated the need for a training on ginger and turmeric juice extractor. The claimed they have been producing taho and turmeric juice but these were all done manually. That is why they badly need the machine especially if they process a large volume of ginger and turmeric. Aside from needing a training on turmeric and ginger extractor, majority of the respondents need a training on band saw operation for their souvenir items, oyster shucker, greenhouse construction, coco shell processor, capiz shell processor, engineered bamboo processor, buko trimmer, coconut dehusker, coconut splitter, coconut decorticator, coconut juicer, coconut scraper, coconut milk extractor, copra dryer, oil expeller, chain saw/ coco lumber saw operation and coconut waste shredder.

Table 3a. Percentage of respondents needing a training on livelihood.

PANGABUHAN SANG PAMILYA (Livelihood)	%	PANGABUHAN SANG PAMILYA (Livelihood)	%
Digital Graphic and Logo Making		Duckery	
Photography	6.67	Native chicken production	66.67
Carpentry	6.67	Native swine production	66.67
Coco Sugar Production	53.33	Basic welding	40.00
Coconut Production	6.67	Organic Farming	53.33
Grouper Culture	6.67	Baking	66.67
Sea bass Culture		Beauty Care	
Tilapia, Catfish, Bangus and Mangroove Crab Production	46.67	Beverages Preparation	
Halo-halo Making	53.33	Candle Making	
Batchoy Making	53.33	Cooking and Cookery	73.33
Virgin Coconut Oil Making	73.33	Doormat and Rag Making	53.33
Cooking Oil Making	60.00	Dress Making	6.67
Handicraft /Souvenir Items Making	53.33	Flower Arrangement	
Vegetable Production	60.00	Fruits and Vegetable Fermentation	
Kakanin Making	73.33	Housekeeping	
Vinegar Making	53.33	Food preservation and processing	53.33
Taho Making		Pasta Noodles Processing	53.33
Salted Egg Production		Hand Soap Making	33.33
Table setting and skirting		T-shirt Printing	13.33
Bottled sardines processing		Bangus Deboning	
Kropek processing		Silag processing	
Siomai Making		Tapa Making	
Chorizo Making		Smoke bangus production	

Table 3b. Percentage of respondents needing a training on Education and Humanities.

EDUKASYON AND HUMANITIES	%	EDUKASYON AND HUMANITIES	%
Pagpanudlo nahanungod sa Edukasyon (Literacy)		Pagpanudlo nahanungod sa paunang lunas (First Aid)	
Pagpanudlo nahanungod sa Ikaayong Lawas (Health)	13.33	Pagpanudlo nahanungod sa ICT	
Pagpanudlo nahanungod bilang ginikanan (Parenting)	20.00	Pagpanudlo nahanungod sa Negosyo (Bookeeping, Marketing, Costing and Accounting)	
Pagpanudlo nahanungod sa katinlo (Hygiene)	13.33	Pagpanudlo nahanungod sa Values Formation	
Pagpanudlo nahanungod sa Culture and Arts		Pagpanudlo nahanungod sa Pagpanalawsaw (Research)	
Pagpanudlo nahanungod sa Customer Service	53.33	Pagpanudlo nahanungod sa Statistics Data Management	
Pagpanudlo nahanungod sa Pagtatap sa kaugalingon (Personality Development)		Pagpanudlo nahanungod sa Temprano nga Pagmamata (Early Pregnancy)	
Pagpanudlo nahanungod sa Kinamatarong kag Pribilihiyo sang Kababaihan (Women's Rights and Privileges)		Pagpanudlo nahanungod sa Pagmaltratar sa Kababayehan kag Kabataan	
Pagpanudlo nahanungod sa Paglikaw kag Pagtapna sang Balatian	26.67	Pagpanudlo nahanungod sa Ladership Management	13.33
Pagpanudlo nahanungod sa 3Rs sang Basura			

Table 3c. Percentage of respondents needing a training on community needs and technology.

KINAHANGLANON SA KOMUNIDAD KAG TEKNOLOHIYA	%	KINAHANGLANON SA KOMUNIDAD KAG TEKNOLOHIYA	%
Coastal Management	6.67	Landscaping and Beautification	13.33
Disaster Preparedness	40.00	Peace and Order	13.33
Fishing Technique	6.67	Solid Waste Management	13.33
Herbal Garden	40.00	Waste Segregation	13.33
Parliamentary Procedure	46.67	Barangay Justice System	33.33
Good Aquaculture Practices	40.00	Animal Nutrition Enhancement Technique and Principles	40.00
Mangroove Rehab and Conservation	40.00	Oyster shucker	66.67
Operation of Aqua&Farm Machineries	33.33	Shell pulverizer/grinder	53.33
Fish Processor	33.33	Band saw	66.67
Corn Mill	33.33	Art materials	53.33
Corn Sheller	13.33	Generator	40.00
Feed mixer	13.33	Diesel and gasoline engine operation	40.00
Rice thresher	6.67	Irrigation Pumps	40.00
Rice cleaner	6.67	Sprayer	33.33
Grain dryer	40.00	Rice Transplanter	33.33
Rice mill	33.33	Rice seeder	33.33
Fish Dryer	53.33	Corn seeder	26.67
Fish smokehouse stove	33.33	Combine harvester	26.67
Coconut/Buko Trimmer	53.33	Electrical wiring installation	26.67
Coconut Dehusker	53.33	Green house construction	53.33
Coconut Grater	53.33	Biogas digester	26.67
Coconut Splitter	53.33	Root crop chipper	40.00
Coconut Decorticator	53.33	Banana chipper	40.00
Coconut Juicer	53.33	Sugarcane juicer	33.33
Coconut Scraper	53.33	Others	
Coconut Milk Extractor	53.33	Turmeric and ginger juice extractor	80.00
Copra dryer	53.33	Coco shell processor, puncher	66.67
Coconut Oil Expeller	53.33	Capiz shell processor, puncher	66.67
Coconut chain saw/coco lumber saw	53.33	Buri processor	46.67
Coconut waste shredder	53.33	Fruit juice extractor	46.67
Briquetting machine	40.00	Engineered bamboo processor	53.33

LESSONS LEARNED

In the engagement of CAPSU and DAR relative to the Coconut Livelihood Program, the following learnings were gained:

- (a) Having Quick Minds During Unanticipated Situations** . There was a problem on the availability of electricity on the second day of the training. It was not anticipated by the CAPSU facilitators nor by the LARC officers. The BSABE Coconut grater, being electronic, could not operate. Good enough the facilitators had a brilliant idea to let the participants do the grating manually while waiting for the electricity to come back. This made them realize how harder it is to grate manually than using the BSABE machine. Also, it made them compare how easy it is to extract coconut milk from grated coconut generated by the BSABE machine than the ones produced manually.
- (b) Differences in Cooking Practices.** It was found out that there is a different way of making *kakanin* (*biko*) in this part of Capiz. In cooking *biko*, the facilitators thought of mixing the brown with coconut milk. However, the participants argued to make use of the coconut oil (after removal of *latik*) instead of coconut milk. The product turned out delicious and less laborious. It turned out that the facilitators learned from the participants a different way of cooking *biko*.
- (c) Amazing Participants and Facilitators.** The participants were a mix of old and young people but managing them was so easy because of their respect, commitment and interest to learn. It was not expected that a lot of them were observant of the processes. An old participant commented on the IEC materials they received and she had a point. In this line, the extension materials being distributed to clientele should be reviewed/checked thoroughly. The participants' place of residence or age can never be underestimated as to the knowledge and skills possessed. Moreover, the facilitators commitment to the program is very high. The distance, summer heat and hardships are nothing as seen in their happy faces and memorable experiences after the extension activities.

RECOMMENDATIONS

For the betterment of the extension program of the school and the clienteles, the following recommendations are hereby offered:

1. CapSU should continue its financial support to the extension programs of the school particularly those lined-up for CAPSU-DAR Coconut Livelihood Program.
2. The BSABE Faculty should engage in collaboration with other agencies like the Philippine Coconut Authority and faculty of other curricular programs for training needs beyond their specialization or capacity.
3. There should be a continuation of the projects and trainings lined up in the Coconut Livelihood Program of CAPSU and DAR-Capiz.
4. This Coconut Livelihood Program be duplicated to beneficiaries of DCAR other than LARC.
5. There should be constant monitoring and maintenance check up of the coconut dehusker and other upcoming coconut technologies by CAPSU BSABE and fabricator/s.
6. There should be impact studies on the trainings conducted.

LITERATURE CITED

Mccormick, E. 2017. Coconut – Tree of Life. <http://studymoose.com/coconut> - tree of life essay

Phillipine Coconut Authority. 2022. <https://www.google.com/search?q=Philippine+coconut+industry>

